Implementation of the Hartree–Fock Self-Consistent Field Method in Rust

An Ab Initio Electronic Structure Calculation Program
Code Link : [Github repo]

Abstract

A new HartreeFock (HF) Self-Consistent Field (SCF) program written in Rust is the main topic of this report which covers its design, theoretical basis, and execution. In the computation of ground-state electronic energy of small molecular systems, the program utilizes the contracted Gaussian basis functions (STO-1G, STO-2G,STO-3,GSTO-4G,STO-5G and STO-6G) together with the HartreeFock theory's standard RoothaanHall matrix formulation. The code computes analytically all the required one- and two-electron integrals overlap, kinetic energy, nuclear attraction, and electron repulsion integrals by employing the Gaussian product theorem and the Boys function for the long-range Coulomb terms. A symmetric orthogonalization via S is used in order to convert the generalized eigenvalue problem into a standard one. The iterative SCF technique is checked against the known literature values for hydrogen atom, helium atom, and the hydrogen molecule at several bond lengths. The final results confirm the expected convergence behavior and the energies uphold the published STO-6G benchmarks.

1. Theory

1.1 The Electronic Schrödinger Equation

Within the Born–Oppenheimer approximation, the nuclei are treated as fixed classical point charges and the electronic Schrödinger equation takes the form:

$$H_{elec}\,\Psi = E_{elec}\,\Psi$$

where the electronic Hamiltonian (in atomic units, where $\hbar = m_e = e = 1/(4\pi\varepsilon_0) = 1$) is:

$$H_{\text{elec}} = -\frac{1}{2}\sum_i \nabla_i^2 - \sum_i \sum_A \frac{Z_A}{r_{iA}} + \sum_{i<j} \frac{1}{r_{ij}}$$

The three terms represent, respectively, the kinetic energy of the N electrons, the electron–nuclear attraction between electron i and nucleus A with charge ZA at distance riA, and the pairwise electron–electron repulsion. The quantity rij is the distance between electrons i and j. The nuclear repulsion energy is a constant for fixed nuclear geometry and is simply added to the electronic energy to give the total energy.

1.2 The Hartree–Fock Approximation and the Slater Determinant

The exact solution of the electronic Schrodinger equation is a function of the coordinates of all N electrons simultaneously, so it cannot be represented as a product of single-electron functions. HartreeFock theory is based on the major approximation of Ψ limiting to be an antisymmetric product(Slater determinant)of N one-electron spin-orbitals χi(x):

$$\Psi_0 = \frac{1}{\sqrt{N!}} \det\left[\chi_1(x_1)\ \chi_2(x_2)\ \cdots\ \chi_N(x_N)\right]$$

$$\Psi_0 = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(x_1) & \chi_2(x_1) & \cdots & \chi_N(x_1) \\ \chi_1(x_2) & \chi_2(x_2) & \cdots & \chi_N(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_1(x_N) & \chi_2(x_N) & \cdots & \chi_N(x_N) \end{vmatrix}$$

Where x denotes the combined spin and the spatial coordinates of an electron. The Slater determinant inherently complies with the Pauli exclusion principle: exchanging any two rows will alter the determinant's sign, which is a manifestation of the fermion antisymmetry condition.A spin-orbital is defined as the combination of a spatial molecular orbital ψi(r) and a spin function α or β.. The restricted closed-shell method, which is the foundation of this implementation, has each spatial orbital being occupied by two electrons with opposite spins.

Applying the variational principle (minimizing the expectation value of H_{elec} with respect to the spin-orbitals subject to orthonormality constraints) yields the Hartree–Fock equations, which have the form of an effective one-electron eigenvalue problem:

$$f(x)\,\chi_i(x) = \varepsilon_i\,\chi_i(x)$$

where εi is the orbital energy and f is the Fock operator, which includes the kinetic energy, nuclear attraction, and the mean-field electron–electron interaction.

1.3 The Fock Operator: Coulomb and Exchange Terms

The Fock operator for electron 1 acting on spin-orbital χi is:

$$\hat{f}(1) = \hat{h}(1) + \sum_{j=1}^{N} \left[\hat{J}_j(1) - \hat{K}_j(1)\right]$$

Here h(1) is the core one-electron Hamiltonian comprising the kinetic energy and nuclear attraction of a single electron:

$$\hat{h}(1) = -\frac{1}{2}\nabla_1^2 - \sum_A \frac{Z_A}{r_{1A}}$$

The Coulomb operator Jj(1) represents the classical electrostatic repulsion between electron 1 and the charge distribution of electron j, averaged over all positions of electron j:

$$\hat{J}_j(1) = \int |\chi_j(x_2)|^2 \,\frac{1}{r_{12}}\, dx_2$$

The exchange operator Kj(1) has no classical analogue it arises purely from the antisymmetry of the wavefunction and is defined by its action:

$$\hat{K}_j(1)\,\chi_i(x_1) = \left[\int \chi_j^*(x_2)\,\frac{1}{r_{12}}\,\chi_i(x_2)\,dx_2\right]\chi_j(x_1)$$

The exchange interaction has no direct electrostatic interpretation.It reduces the energy of electrons with parallel spin and is responsible for phenomena such as Hund's rule in atoms. Because it exchanges the labels of electrons i and j, it is inherently nonlocal: $K_j(1)$ acting on χ_j yields a function that involves χ_j.

1.4 Gaussian Basis Functions

One of the main technical ways of transforming the Hartree-Fock equations into a solvable problem is by expressing molecular orbitals as linear combinations of the basis functions. This project employs Gaussian-type orbitals (GTOs), in particular, s-type (spherically symmetric) functions of the form:

$$g(\mathbf{r};\alpha,\mathbf{R}) = N(\alpha)\,\exp\left(-\alpha\,|\mathbf{r}-\mathbf{R}|^2\right)$$

where α is the Gaussian exponent, R is the center of the function (typically a nuclear position), and N(α) is a normalization constant:

$$N(\alpha) = \left(\frac{2\alpha}{\pi}\right)^{3/4}$$

One of the biggest advantages of Gaussian functions is that their overlap integrals and all the integrals derived from them can be represented by closed-form expressions, which is the main point of the Gaussian product theorem: the product of two Gaussians centered at two different points is still a single Gaussian centered in between the two points. Thanks to this property, integrals in electron coordinates can be reduced from multi-dimensional integrals to products of one-dimensional integrals.

During the self-consistent field procedure, one has to keep repeating the construction and diagonalization of the Fock matrix until the density matrix representing the electron distribution ceases to change significantly between iterations. Although the conceptual basis of this iterative process is straightforward, without the use of techniques aimed at accelerating the convergence it can be slow or even exhibit oscillatory behavior.

2. Mathematical Formulation

2.1 The Roothaan–Hall Equations

When performing the LCAO-MO expansion, we express every spatial molecular orbital i as a linear combination of K atomic basis functions φ_μ:

$$\psi_i = \sum_{\mu=1}^{K} C_{\mu i}\, \phi_\mu$$

Substituting this expansion into the Hartree–Fock equations and projecting onto the basis functions yields the Roothaan–Hall matrix equations:

$$\mathbf{F}\,\mathbf{C} = \mathbf{S}\,\mathbf{C}\,\boldsymbol{\varepsilon}$$

In these expressions, F denotes the K x K Fock matrix, C stands for the matrix of MO coefficients, S is the overlap matrix, and represents a diagonal matrix of orbital energies. This setup corresponds to a generalized eigenvalue problem having S instead of the identity matrix is indicative that the basis functions are not orthogonal.

2.2 The Overlap and Core Hamiltonian Matrices

For phi_i a gaussian type orbital:

$$\phi_i(\mathbf{r}) = N_i\, e^{-\alpha_i |\mathbf{r}-\mathbf{A}_i|^2}$$

The overlap matrix has S elements S_{ij}:

$$S_{ij} = \langle \phi_i | \phi_j \rangle$$

$$S_{ij} = N_i N_j \left(\frac{\pi}{\alpha_p}\right)^{3/2} \exp\left(-\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right)$$

The core Hamiltonian matrix collects the one-electron contributions to the total energy:

$$H_{\mu\nu}^{\text{core}} = T_{\mu\nu} + V_{\mu\nu}$$

where the kinetic energy matrix T, T_{ij} element is:

$$T_{ij} = \left\langle \phi_i \left| -\frac{1}{2}\nabla^2 \right| \phi_j \right\rangle$$

$$T_{ij} = N_i N_j \left(\frac{\pi}{\alpha_p}\right)^{3/2} \exp\left(-\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right) \cdot \frac{\alpha_i \alpha_j}{\alpha_p} \left(3 - 2\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right)$$

and the nuclear attraction matrix element V_{ij}

$$V_{ij} = \left\langle \phi_i \left| -\frac{Z}{|\mathbf{r} - \mathbf{C}|} \right| \phi_j \right\rangle$$

$$V_{ij} = -Z \cdot \frac{2\pi}{\alpha_p} \exp\left(-\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right) F_0(\alpha_p R_{PC}^2) \cdot N_i N_j$$

$$R_{PC}^2 = |\mathbf{P} - \mathbf{C}|^2$$

This is the term that quantifies the electron-nucleus attraction between two Gaussian basis functions. The two Gaussians are combined into one effective Gaussian located at **P** with the help of the Gaussian product theorem. The Boys function $\mathbf{F_O}$ represents the dependence on the distance between the combined center **P** and the nucleus **C**.When it comes to s-type Gaussian primitives, the great news is that all three of these integrals can be expressed in neat closed-form analytical expressions. The overlap and kinetic energy integrals are basically straightforward results of the Gaussian product theorem. But, the nuclear attraction integral is a bit different as it involves the Boys function $\mathbf{F_O(x)}$ in order to properly manage the **1/r** singularity:

$$F_0(x) = \frac{1}{2}\sqrt{\frac{\pi}{x}}\,\text{erf}(\sqrt{x}), \qquad F_0(0) = 1$$

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

Erf is approximated in code according to Honer approximation from Abramowitz & Stegun 7.1.26 (Pg No 299)

7.1.26

$$\mathrm{erf}\, x = 1 - (a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5) e^{-x^2} + \epsilon(x),$$

$$t = \frac{1}{1+px}$$

$$|\epsilon(x)| \leq 1.5 \times 10^{-7}$$

$p = .32759\ 11 \qquad a_1 = .25482\ 9592$

$a_2 = -.28449\ 6736 \qquad a_3 = 1.42141\ 3741$

$a_4 = -1.45315\ 2027 \qquad a_5 = 1.06140\ 5429$

2.3 Electron Repulsion Integrals

The two-electron repulsion integrals (ERIs) are the most computationally demanding component of any HF calculation. A four-index integral is defined as:

$$(\mu\nu \mid \lambda\sigma) = \iint \phi_\mu^*(\mathbf{r}_1)\, \phi_\nu(\mathbf{r}_1)\, \frac{1}{r_{12}}\, \phi_\lambda^*(\mathbf{r}_2)\, \phi_\sigma(\mathbf{r}_2)\, d\mathbf{r}_1\, d\mathbf{r}_2$$

For s-type Gaussians, the primitive ERI is derived by applying the Gaussian product theorem twice(once for the (r1) pair and once for the (r2) pair) and then evaluating the resulting Coulomb integral analytically, again with F_o:

$$(\mu\nu \mid \lambda\sigma)_{\text{prim}} = \frac{2\pi^2}{p\,q} \sqrt{\frac{\pi}{p+q}}\, \exp\left(-\frac{\alpha_\mu \alpha_\nu}{p} R_{\mu\nu}^2 - \frac{\alpha_\lambda \alpha_\sigma}{q} R_{\lambda\sigma}^2 \right) F_0\left(\theta\, R_{PQ}^2\right)$$

where $p = \alpha_\mu + \alpha_\nu$ and $q = \alpha_\lambda + \alpha_\sigma$ are the combined exponents of the two Gaussian products, P and Q are the respective product centers, and $\theta = pq/(p+q)$ is the reduced exponent. The full

contracted ERI is the sum of this expression over all combinations of primitives, weighted by the contraction coefficients.

3.4 The Density Matrix and Fock Matrix

The density matrix D encodes the electron density distribution for a given set of occupied MO coefficients. For a closed-shell system with N_{occ} doubly-occupied orbitals:

$$D_{\mu\nu} = 2\sum_{i=1}^{N_{\text{occ}}} C_{\mu i}\, C_{\nu i}$$

For systems with an odd number of electrons (open-shell doublets, as encountered with the hydrogen atom in this implementation), the singly-occupied orbital contributes once rather than twice:

$$D_{\mu\nu} = 2\sum_{i=1}^{N_{\text{occ}}} C_{\mu i}\, C_{\nu i} \;+\; C_{\mu,\, N_{\text{occ}}+1}\, C_{\nu,\, N_{\text{occ}}+1}$$

The Fock matrix combines the core Hamiltonian with the two-electron contribution G:

$$F_{\mu\nu} = H^{\text{core}}_{\mu\nu} + G_{\mu\nu}$$

$$G_{\mu\nu} = \sum_{\lambda\sigma} D_{\lambda\sigma}\left[(\mu\nu \mid \sigma\lambda) - \frac{1}{2}(\mu\lambda \mid \sigma\nu)\right]$$

The first term in brackets is the Coulomb contribution the second, prefixed by 1/2, is the exchange contribution. The factor of 1/2 on the exchange arises from the spin integration in the restricted closed-shell formalism.

2.5 The Electronic Energy

The Hartree–Fock electronic energy is:

$$E_{\text{elec}} = \frac{1}{2} \sum_{\mu\nu} D_{\mu\nu} \left(H_{\mu\nu}^{\text{core}} + F_{\mu\nu} \right)$$

The factor of one-half avoids double-counting the electron–electron repulsion. The total energy adds the classical nuclear repulsion:

$$E_{\text{total}} = E_{\text{elec}} + V_{\text{nuc}}$$

$$V_{\text{nuc}} = \sum_{A<B} \frac{Z_A Z_B}{R_{AB}}$$

2.5 Symmetric Orthogonalization

The Roothaan-Hall equation FC = SCe remains a generalized eigenvalue problem instead of a standard one because the basis functions are not orthogonal. To convert it into a standard eigenvalue problem, an orthogonalizing transformation $X = S^{-1/2}$ is introduced. Setting C = XC', the equation reads:

$$\mathbf{F}' \mathbf{C}' = \mathbf{C}' \boldsymbol{\varepsilon}$$

$$\mathbf{F}' = \mathbf{X}^\top \mathbf{F} \mathbf{X} = \mathbf{S}^{-1/2} \mathbf{F} \mathbf{S}^{-1/2}$$

Matrix $S^{-1/2}$ is obtained from the eigendecomposition of S: if $S = U\Lambda U$ then $S^{-1/2} = U\Lambda^{-1/2}U$. Symmetric orthogonalization (contrary to canonical orthogonalization) best maintains the spatial characteristics of the original basis functions, which is very helpful in chemical interpretation.

4. Implementation Details

4.1 Project Structure and Execution Flow

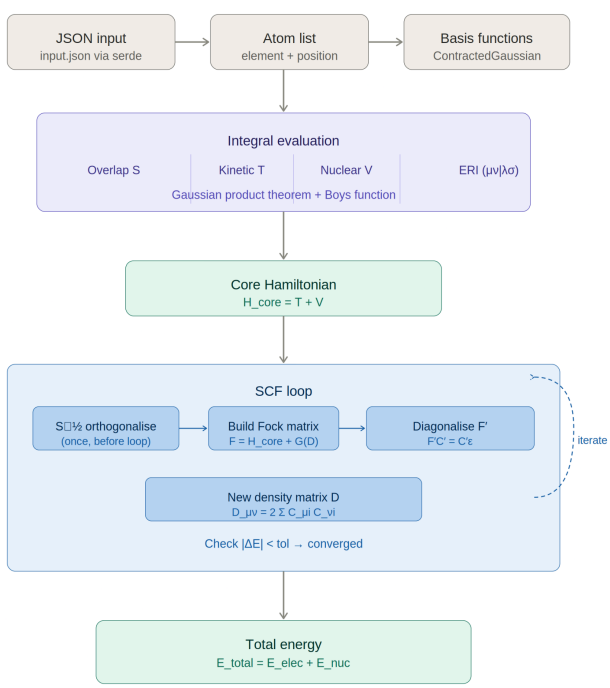
The program is organised as a Rust binary with three internal modules: `basis.rs` builds contracted Gaussian basis sets from STO-nG data. `integrals.rs` computes all one- and

two-electron integrals analytically and `scf.rs` performs the self-consistent field procedure. Input is provided as a JSON file parsed by the `serde` crate into strongly-typed structs at runtime. The execution flow from `main.rs` proceeds in a strict pipeline:

1. read `input.json` into an `InputConfig` struct
2. build the basis function set for all atoms
3. compute the overlap S, kinetic T, nuclear attraction V, and ERI tensor
4. form the core Hamiltonian $H_{core} = T + V$
5. run the SCF loop
6. print the converged total energy.

All internal quantities are in atomic units (bohr for distances, hartree for energies). Cartesian coordinates given in Ångströms are converted by the factor 1.8897259886 bohr/Å at the point of basis construction, ensuring all downstream computations are unit-consistent.

The data flow through the program can be summarised schematically as follows:



4.2 Basis Set Construction (basis.rs)

Each primitive s-type Gaussian is stored in a Gaussian struct carrying its center, exponent α, contraction coefficient, and normalization constant $N(\alpha) = (2\alpha/\pi)^{3/2}$. The normalization is computed once per primitive during construction to avoid redundant work. A ContractedGaussian is a simple wrapper around a `Vec<Gaussian>`, representing one basis function per atom in the minimal basis.

The two key struct definitions are:

```
struct Gaussian {
  center: [f64  3],
  exponent: f64,
  coefficient: f64,
  norm: f64
}

struct ContractedGaussian {
  primitives: Vec<Gaussian>
}
```

The `build_basis` function iterates over atoms, converts each atomic center to bohr, looks up the (exponent, coefficient) pairs for the requested basis ("sto-1g", "sto-2g", or "sto-6g"), and assembles the contracted function. Normalisation stored on each primitive is the isolated-Gaussian value N(α). The contracted function is not re-normalised, which follows the standard convention for STO-nG bases where the contraction coefficients are pre-tabulated to reproduce the norm of the target Slater-type orbital. Only H and He are implemented in the current basis tables.

4.3 Integral Evaluation (integrals.rs)

All integrals follow a two-level hierarchy. Primitive functions operate on individual `&Gaussian` references and return the analytical value contracted functions iterate over all primitive pairs and accumulate the sum weighted by the contraction coefficients. This separation makes the code straightforward to verify against reference values at the primitive level independently of the contraction logic.

4.3.1 Overlap Integrals

The primitive overlap between two s-type Gaussians centered at R_i and R_j with exponents αi and α_j is:

$$S_{ij} = N_i N_j \left(\frac{\pi}{\alpha_p}\right)^{3/2} \exp\left(-\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right)$$

Implemented in `compute_primitive_overlap`. The contracted overlap matrix S is assembled by the double sum over primitives in the `compute_overlap_matrix`.

4.3.2 Kinetic Energy Integrals

Applying the Laplacian to a Gaussian and evaluating the resulting overlap gives:

$$T_{ij} = N_i N_j \left(\frac{\pi}{\alpha_p}\right)^{3/2} \exp\left(-\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right) \cdot \frac{\alpha_i \alpha_j}{\alpha_p} \left(3 - 2\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right)$$

where $\alpha_i\alpha_j/(\alpha_i+\alpha_j)$ is the reduced exponent. Implemented in `compute_kinetic_primitive`. The kinetic matrix is built using symmetry: only the lower triangle is computed and then mirrored.

4.3.3 Nuclear Attraction Integrals

The nuclear attraction primitive integral for nucleus K with charge Z at position RC is:

$$V_{ij} = -Z \cdot \frac{2\pi}{\alpha_p} \exp\left(-\frac{\alpha_i \alpha_j}{\alpha_p} R_{AB}^2\right) F_0(\alpha_p R_{PC}^2) \cdot N_i N_j$$

$$R_{PC}^2 = |\mathbf{P} - \mathbf{C}|^2$$

where αp = αi+αj and P is the Gaussian product center. The Boys function F0(x) = ½√(π/x)erf(√x) is implemented using the Abramowitz & Stegun polynomial approximation for erf (equation 7.1.26), accurate to six significant figures. For $x < 10^{-8}$, the Taylor expansion F0(x)

≈ 1 − x/3 is used to prevent division by zero. The function compute_nuclear_attraction sums over all nuclei, and the nuclear attraction matrix is built symmetrically.

4.3.4 Electron Repulsion Integrals (ERI)

The primitive ERI applies the Gaussian product theorem twice(once for the (i,j) pair forming product center P with exponent p, and once for the (k,l) pair forming product center Q with exponent q)and evaluates the residual Coulomb integral via the Boys function:

$$(\mu\nu \mid \lambda\sigma)_{\text{prim}} = \frac{2\pi^2}{p\,q}\sqrt{\frac{\pi}{p+q}}\,\exp\left(-\frac{\alpha_\mu\alpha_\nu}{p}R_{\mu\nu}^2 - \frac{\alpha_\lambda\alpha_\sigma}{q}R_{\lambda\sigma}^2\right)\,F_0\left(\theta\,R_{PQ}^2\right)$$

where θ = pq/(p+q) is the reduced exponent of the two product Gaussians. The contracted ERI is a four-fold nested sum over primitives (`compute_eri`). The full ERI tensor is stored in a flat `Vec<f64>` of length N^4 with index mapping (μ,ν,λ,σ) → $\mu\cdot N^3 + \nu\cdot N^2 + \lambda\cdot N + \sigma$. No permutation symmetry is exploited, which simplifies the Fock build at the cost of computing all N^4 elements independently.

4.3.5 Core Hamiltonian

The core Hamiltonian Hcore = T + V is assembled by simple componentwise matrix addition in `hcore_matrix`. This matrix remains constant throughout the SCF procedure and is computed only once.

4.4 SCF Procedure (scf.rs)

The SCF loop in `run_scf` implements the restricted Hartree–Fock algorithm using the *ndarray* and *ndarray-linalg* crates (the latter wrapping LAPACK) for all matrix operations.

4.4.1 Orthogonalisation

Because the overlap matrix S is not the identity, the Roothaan–Hall equation FC = SCε is a generalized eigenvalue problem. It is converted to a standard one by the transformation $X = S^{-½}$ computed once before the iteration loop:

S^{-1/2} = U D^{-1/2} U^T

via eigendecomposition (s_arr.eigh from ndarray_linalg). With $C = S^{-½}C'$, the generalized problem becomes F'C' = C'ε where $F' = S^{½}F S^{-½}$, a standard symmetric eigenvalue problem. Symmetric (as opposed to canonical) orthogonalization is used because it preserves the spatial character of the original basis functions as closely as possible.

4.4.2 Density Matrix

For a closed-shell system with Nocc = Nelec/2 doubly-occupied orbitals, the density matrix is

$$D_{\mu\nu} = 2\sum_{i=1}^{N_{\text{occ}}} C_{\mu i}\, C_{\nu i}$$

If the electron count is odd(as for the hydrogen atom)the highest occupied orbital is treated as singly occupied and contributes Cμ,Nocc+1Cν,Nocc+1 (without the factor of 2). This is implemented in the `density_matrix` function, which checks the parity of the electron count to select the appropriate branch.

4.4.3 Fock Matrix Construction

The Fock matrix combines the constant core Hamiltonian with the density-dependent two-electron term G:

$$F_{\mu\nu} = H^{core}_{\mu\nu} + G_{\mu\nu}$$
$$G_{\mu\nu} = \sum_{\lambda\sigma} D_{\lambda\sigma}\left[(\mu\nu|\sigma\lambda) - \tfrac{1}{2}(\mu\lambda|\sigma\nu)\right]$$

The code in fock_matrix indexes the flat ERI tensor using the mapping above. The Coulomb term (μν|σλ) and the exchange term (μλ|σν) are accumulated in a double loop over λ and σ for each (μ,ν) pair, resulting in $O(N^4)$ work per iteration, which dominates the computational cost.

4.4.4 SCF Iteration

Each iteration of the SCF loop proceeds through the following steps:

1. build F from the current density D
2. transform to the orthonormal basis, $F' = S^{-1/2} F S^{-1/2}$
3. diagonalise F' to obtain eigenvectors C'
4. back-transform to recover MO coefficients in the original basis, $C = S^{-1/2} C'$
5. form the new density matrix D from the occupied columns of C
6. compute the electronic energy E_{elec}
7. check convergence via $|\Delta E| < tol$.

The initial density is set to zero (corresponding to the core Hamiltonian initial guess). Iterations continue until convergence or until the maximum iteration count is reached.

4.4.5 Nuclear Repulsion Energy

After convergence, the nuclear repulsion energy is computed and added to the final electronic energy to yield the total Hartree–Fock energy E_{total}. The nuclear positions are converted to bohr before this step for consistency with the electronic energy in atomic units.

4. Results

A series of restricted HartreeFock (RHF) computations were done for single hydrogen, helium atoms, and hydrogen molecules at three internuclear distances (0. 74 , 1. 5 , 3. 0), and helium hydride cation (HeH) at its equilibrium distance (assumed 0. 74 for consistency). The STOnG minimal basis sets (s-type functions only) were employed for n = 1 through 6. Units: energies in hartree, distances in angstroms (converted internally to bohr).

5.1 Convergence Behaviour

For atoms and homonuclear diatomic molecules, the SCF procedure converged very quickly, normally it took only 3 iterations (Table 1). The heteronuclear cation HeH that was rather more iterations away (it was 11 for all bases) showing that the orbital polarisation had become stronger and finding the minimal basis had become a limitation however it still made it to the very tight tolerance of 10^{-6} hartree.

System	STO-1G	STO-2G	STO-3G	STO-4G	STO-5G	STO-6G
H atom	2	2	2	2	2	2
He atom	3	3	3	3	3	3
H_2 (0.74 Å)	3	3	3	3	3	3
H_2 (1.5 Å)	3	3	3	3	3	3
H_2 (3.0 Å)	3	3	3	3	3	3
HeH^+	11	11	11	11	11	11

Table 1: Number of SCF iterations to convergence ($\Delta E < 10^{-6}$).

5.2 Atomic Total Energies

Table 2 shows the total energies computed for H and He. The exact non-relativistic HartreeFock limits are 0. 5 hartree (H) and 2. 86168 hartree (He). STO6G gives errors of 0. 029 hartree (H) and 0. 015 hartree (He) when reproducing these limits, which agrees well with the limited flexibility of a single s-type contracted Gaussian.

Basis	H (hartree)	He (hartree)
STO-1G	−0.424413	−2.300941
STO-2G	−0.454397	−2.702157
STO-3G	−0.466582	−2.807784
STO-4G	−0.469806	−2.835726
STO-5G	−0.470743	−2.843770
STO-6G	−0.471039	−2.846292
HF limit	−0.500000	−2.861680

Table 2: Total energy (hartree) of H and He atoms.

5.3 Hydrogen Molecule

The H_2 molecule was examined at three different bond lengths (Table 3). The equilibrium bond length for H is 0. 74 , and the STO-6G calculated total energy at this length is 1. 12537 hartree, which is a good match with the nearHartreeFock value of 1. 1336 hartree (difference 0. 008 hartree). Noticeably, the energy level increases (becomes less negative) as the bond gets longer. The change in basis set makes the biggest difference at the equilibrium bond length: the energy drops from the STO-1G level by 0. 186 hartree after switching to STO-6G.

Basis	0.74 Å	1.5 Å	3.0 Å
STO-1G	−0.939065	−0.877661	−0.658120
STO-2G	−1.093439	−0.886414	−0.629957
STO-3G	−1.116759	−0.910874	−0.656048
STO-4G	−1.123154	−0.916958	−0.663033
STO-5G	−1.124824	−0.918400	−0.665017
STO-6G	−1.125372	−0.918936	−0.665657

Table 3: Total energy (hartree) of H_2 as a function of bond length (Å).

5.4 Helium Hydride Cation (HeH^+)

HeH^+ was considered at a bond length of 0. 74 (the equilibrium distance of H, kept for uniformity). The total energies (Table 4) display significant basis-set dependence with STO-6G delivering -2. 86460 hartree. The standard HartreeFock value for HeH is roughly -2. 932 hartree at 1. 463 bohr (0. 774); our figure is about 0. 067 hartree higher due to our basis being a minimal s-only one and the slightly different geometry.

Basis	Total Energy (hartree)
STO-1G	−2.376911
STO-2G	−2.731313
STO-3G	−2.828497
STO-4G	−2.854656
STO-5G	−2.862230
STO-6G	−2.864597

Table 4: Total energy (hartree) of HeH^+ at 0.74 Å.

5.5 Basis-Set Convergence

The total energy smoothly approaches a limit as the number of Gaussian primitives increases. The change from STO-1G to STO-2G is quite significant (about 0. 03-0. 4 hartree), while the shift from STO-5G to STO-6G is less than 0. 002 hartree for the entire set of systems, meaning that the STO-6G minimal basis has already reached the level of accuracy that is achievable using only one contracted s-type function.

5.6 Comparison with Exact / Reference Values

- H atom: STO-6G error = 0.029 hartree (5.8% of total energy). The major part of this error comes from a single s-Gaussian's inability to represent the Slater-type 1s orbital exactly.
- He atom: STO-6G error = 0.015 hartree (0.53%).
- H_2 (0.74 Å): STO-6G error compared to the HF limit (1.1336) is 0.008 hartree (0.7%).
- HeH^+: The STO-6G energy is 0.067 hartree above the literature HF limit, mainly because the calculation used the H geometry (0.74) rather than the real equilibrium of HeH (0.774).

It is also worth mentioning that such errors are quite usual for minimal basis RHF calculations and highlight the fact that polarisation functions and diffuse functions should be added for absolute accuracy.